

Initial Report

Multi-Drone Exploration with Random Faults

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Abstract

This project aims to design and implement a cooperative multi-agent system for drone exploration in a grid-world environment. A team of drones must search for a hidden target under partial observability, limited communication, and a fixed step budget. To make the task even harder we introduce random drone faults: during an episode, one or more drones may break and become inactive, forcing the remaining agents to adapt to an unexpected loss of team members. This makes the scenario more suitable for a Multi-Agent Systems project because it highlights cooperation, robustness, and resilience under uncertainty. The expected outcome is a complete software artifact together with a concise final report describing the design choices and experimental results.

Description

The project will focus on building a multi-drone exploration environment from scratch. Each drone will observe only a portion of the grid, exchange information with nearby teammates, and cooperate to find the target as efficiently as possible. The system will be developed as a reinforcement learning project, with a shared policy controlling all drones and a configuration-driven setup for environment and training parameters.

During an episode, each drone may randomly fail and stop acting for the rest of the run. This will allow the analysis of how the system behaves when the team becomes smaller unexpectedly, which is a relevant aspect in multi-agent systems. The step limit will remain active, so the exploration task will depend both on time pressure and on the possibility of losing active agents.

Goals

The project has the following goals:

- design a cooperative multi-agent grid environment for drone exploration;
- implement communication and map sharing among nearby drones;
- train a reinforcement learning policy for cooperative target search;
- introduce random faults that can disable drones during an episode;
- evaluate how the system adapts to unexpected agent loss.

Expected deliverables

The expected deliverables are:

- the source code of the multi-drone exploration system;
- the training and evaluation pipeline for the reinforcement learning agent;
- a final report describing architecture, design choices, and results;
- a GitLab repository containing the complete project material.

Main configurable parameters

The project will include several configurable aspects that can be adjusted during development and experimentation. In particular, the environment will allow the definition of the grid structure, the number of drones, the obstacle density, the sensing radius, the communication range, the movement budget, and the random seed used for reproducibility.

The reward function will also be configurable, so that it is possible to tune the penalties associated with invalid moves, revisiting already explored cells, and collisions among drones, as well as the bonus obtained when new areas are discovered and the terminal reward granted when the target is found.

In addition, the project will include parameters for domain randomization, in order to vary the sensing range, communication range, and team size across episodes. On the learning side, the training process will expose the usual hyperparameters of reinforcement learning, including the discount factor, the return horizon, the replay buffer size, the minibatch size, the target-network update frequency, the update frequency, the gradient clipping threshold, the exploration schedule, and the learning-rate schedule.

These parameters will make it possible to run controlled experiments and compare different cooperative and fault-tolerant behaviours under a variety of settings.

Fault model

A relevant component of the project concerns the modelling of unexpected drone failures during execution. During an episode, one or more drones may become inactive, so the remaining agents must continue the exploration task with reduced operational capacity. This choice makes the scenario more realistic and more suitable for a Multi-Agent Systems perspective, because it introduces fault tolerance, adaptation, and resilience as central aspects of the system rather than marginal ones.

The project will study how cooperative behaviour changes when the team size is reduced unexpectedly, and whether the learned policy is able to maintain effective exploration and target search despite the loss of active agents. This aspect is particularly important because the system must react not only to partial observability and communication constraints, but also to unforeseen disruptions affecting the agents.

Conclusion

This project combines environment design, multi-agent cooperation, reinforcement learning, and simulation within a single framework. Its goal is to develop a cooperative drone exploration system capable of operating under uncertainty, partial information, and possible agent failures. The final report will describe the implemented system, the main design choices, the simulation setting, and the experimental results obtained during the project.